

	or antinode is 10.2 cm above water surface	A1	[2]
7	(a) lamp C lamp is shorted	M1 A1	[2]
	(b) shorted <u>lamp A</u> would cause damage to the supply/lamps /blow fuse in supply	B1	[1]
	(c) $15\ \Omega$	B1	[1]
	(d) (i) $V = IR$ $R = 30\ \Omega$	C1 A1	[2]
	(ii) $P = VI$ or I^2R or V^2/R $P = 1.2\ \text{W}$	C1 A1	[2]
	(e) filament is cold when measuring with ohm-meter in (b) resistance of filament rises as temperature rises	B1 B1	[2]
8	(a) nucleus emits	M1	